

Fertility and Housing Tenure Choice

Tommaso De Santo

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Since May

One-market lifecycle model of housing and fertility. What changed:

- **Sequential fertility:** birth-by-birth attempts under declining fecundity; children ever born and children at home tracked separately
- **Idiosyncratic income risk:** persistent after-tax earnings process, set externally
- **One housing market:** the two-location structure is set aside for the national question

Model

Environment

- One national housing market; stationary equilibrium with household population S
- Households live J four-year periods from age 18; retirement, then survival risk
- Two child states: n children ever born, m children currently at home
 - Children at home raise consumption and housing needs
 - Each child matures stochastically, leaves the household, and enters the economy as a young adult
- Idiosyncratic persistent earnings risk; wages and the interest rate are exogenous
- Households rent, or own a discrete house size on a ladder financed with a mortgage
- Government: property tax rebated lump sum; payroll tax funds a pension
- New households enter as young renters: matured children plus an outside inflow

Preferences

CRRA utility per equivalent adult over a consumption–housing bundle:

$$u(c, \mathbf{h}; m) = \frac{(x/e(m))^{1-\sigma}}{1-\sigma} + \psi_{\text{child}} m, \quad x = c^\alpha (\mathbf{h} - \underline{h} m)^{1-\alpha}.$$

Equivalence scale (Citro–Michael 1995; Scholz–Seshadri–Khitatrakun 2006):

$$e(m) = \left(\frac{2 + 0.7 m}{2} \right)^{0.7}$$

relative to a childless couple: each child at home raises the resources needed to reach the same utility.

Space requirement. Each child at home absorbs $\underline{h}$ rooms before housing yields services.

Housing services. Owners receive a service premium:

$$\mathbf{h} = \mathbf{1}\{\text{own}\} \chi h + \mathbf{1}\{\text{rent}\} h^R, \quad \chi > 1.$$

Bequests

Warm-glow utility at death over terminal wealth:

$$\mathcal{B}(\tilde{b}) = \theta_0 \frac{(\theta_1 + \tilde{b})^{1-\sigma}}{1-\sigma}, \quad \tilde{b} = b + ph.$$

- θ_0 governs the strength of the bequest motive; θ_1 makes bequests a luxury
- Estates are valued by parents but transfer to no one: no intergenerational wealth flow

Fertility and Child Maturation

Between ages 18 and 45, households decide each period whether to try for a child — a discrete choice with taste shocks, with separate scales κ_f before and after the first child.

Trying yields a birth with the conception probability

$$\pi(\text{age}) = 1 - \omega_1 e^{\omega_2(\text{age}-18)}, \quad \pi = 0 \text{ after } 45,$$

which falls from 0.98 at 18 to 0.62 at 40: postponed births are increasingly likely never to happen.

Each child lives at home for 18 years on average (stochastic maturation), raising the equivalence scale and the space requirement; matured children enter the economy as young households.

Earnings and Institutions

Idiosyncratic persistent earnings risk over the lifecycle:

$$\log y_{j+1} = \rho \log y_j + \varepsilon_{j+1}, \quad \varepsilon \sim N(0, \sigma_\varepsilon^2),$$

discretized and combined with an age profile of earnings.

- Earnings are after tax: a progressive retention function applied to gross earnings
- Retirees receive a stationary pension funded by a payroll tax
- Property tax on owners, rebated as a lump-sum transfer

Housing, Tenure, and Budget Constraints

Households rent a continuous quantity up to a cap, or own a discrete size on a ladder:

$$h^R \in (0, \bar{h}^R], \quad h \in \{H_1, \dots, H_K\}.$$

Tenure segmentation: $\max H_k > \bar{h}^R$, so family-sized space is reachable only through ownership.

Renters

$$c + b' + r h^R = (1 + q) b + y(j), \quad b' \geq 0.$$

Owners, with prior stock h_{-1}

$$c + b' + (\delta + \tau_H) ph + \mathbf{1}\{h \neq h_{-1}\} [ph - (1 - \psi) ph_{-1}] = (1 + q) b + y(j), \quad b' \geq -\phi ph.$$

- q : interest rate; δ : maintenance; τ_H : property tax; ψ : transaction cost; $1 - \phi$: down-payment share

Housing Supply and Market Clearing

Housing supply slopes upward in levels, and rents map into prices through user cost:

$$H^S = H_0 \left(\frac{r}{\bar{r}} \right)^\xi, \quad r = (q + \delta + \tau_H) p.$$

- One market-clearing condition: aggregate housing services demanded equal H^S
- Supply is in levels, not per household: a permanently larger population permanently raises rents
- Stationary equilibrium: the price, the transfer, and the distribution are constant, and the government budget clears

The Household's Problem

Within the period, a household first decides whether to try for a child, then chooses tenure, housing, and saving:

$$V_j(b, h, y, n, m) = \mathbb{E}_\varepsilon \max_{a \in \{0,1\}} \left\{ W_j^a(b, h, y, n, m) + \varepsilon_a \right\},$$

$$W_j^a = \max_{\text{rent/own}, h', h^{R'}, b'} u(c, \mathbf{h}; m) + \beta \mathbb{E}[V_{j+1}(b', h', y', n', m')]$$

subject to the renter or owner budget and the collateral constraint; a successful try ($a = 1$, probability π_j) adds a child next period.

In old age there are no fertility decisions; survival risk and the warm glow at death close the problem.

Calibration

Externally Calibrated Parameters

	Object	Value	Source
<i>Pref.</i>	σ risk aversion	2.0	standard
<i>Fin.</i>	q interest (annual)	0.04	standard
	δ depreciation (annual)	0.02	Greaney et al.
	τ_H property tax (annual)	0.01	US average
	ϕ financed share	0.80	20% down payment
	ψ transaction cost	0.06	Greaney et al.
<i>Life</i>	17 four-year periods	ages 18–85	retirement 65
<i>Income</i>	$\rho, \sigma_\varepsilon, \text{HSV } \tau$	0.914, 0.169, 0.181	FL $\times$ HSV
	τ_{pay} payroll	0.179	Greaney et al.
<i>Children</i>	maturation hazard	2/9 per period	18 expected years
	fecundity ω_1, ω_2	0.02, 0.134	conception schedule
<i>Housing</i>	ladder, rental cap	$\{2, \dots, 11\}$, 6 rooms	room units

External restriction on the search: annual $\beta \geq 0.94$.

Internally Calibrated Parameters

Param	Description	Estimate	Bounds	Near bound?
β (annual)	Discount factor	0.9946	[0.94, 0.9995]	no
ψ_{child}	Child flow utility	0.000	[0, 3]	at lower
κ_f	Attempt taste scale, first birth	2.987	[0.02, 50]	no
κ_f^{cont}	Attempt taste scale, later births	1.713	[0.02, 50]	no
χ	Owner service premium	1.037	[0.1, 5]	no
H_0	Housing supply intercept	9.873	[0.2, 80]	no
θ_0	Bequest strength	0.159	[0, 8]	near lower
θ_1	Bequest shifter	0.094	[0.02, 16]	near lower
$\underline{h}$	Child room floor	0.614	[0.1, 1.8]	no

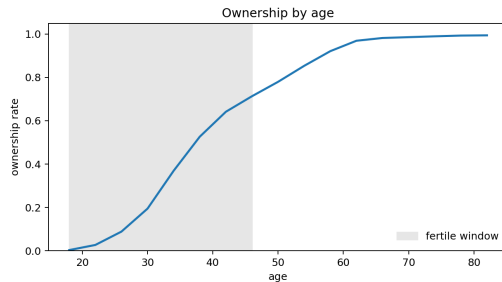
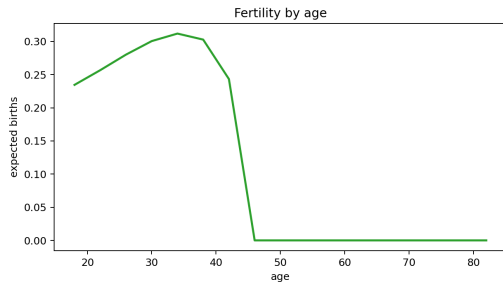
Nine parameters, twelve moments. ψ_{child} is estimated at zero, and the bequest parameters sit near their lower bounds.

Model Fit

	Moment	Target	Model	Gap	Contr.
<i>Fertility</i>	Completed fertility	1.918	1.967	+0.049	3.4
	Childlessness	0.188	0.104	-0.084	121.5
	Mean age at first birth	25.31	25.57	+0.26	1.1
	First births at 30+	0.270	0.327	+0.057	51.5
<i>Housing</i>	Rooms response to first birth	0.664	0.690	+0.026	0.6
	Rooms, 3+ vs. 1-2 children	0.368	0.359	-0.009	0.2
	Family ownership gap	0.168	0.162	-0.006	0.5
	Ownership rate	0.575	0.582	+0.007	0.1
	Aggregate mean rooms	5.780	6.478	+0.698	5.8
<i>Wealth</i>	Wealth / earnings	6.873	5.794	-1.079	7.3
	Bequest flow / wealth	0.0088	0.0087	-0.0001	0.0
	Old wealth p90/p50	3.448	2.090	-1.358	105.1

The remaining misses are childlessness, the dispersion of old-age wealth, and the share of late first births; additional search does not improve them.

Lifecycle Profiles



Population Closure

Demographics and Population Balance

Completed fertility is below replacement: matured children replace about 86% of exiting households. A stationary population therefore needs an inflow from outside — as in the U.S., where net immigration now sustains population growth.

Entrants each period are retained matured children plus an outside inflow M :

$$\underbrace{S E_0}_{\text{entrants needed}} = \underbrace{R S B_0}_{\text{retained matured children}} + M,$$

with E_0 the replacement flow per household, B_0 the matured-children flow, and R the share who stay.

Given a fixed inflow, the stationary population

$$S = \frac{M}{E_0 - R B_0}$$

exists and is stable: as the population shrinks, a fixed inflow becomes a growing share, and the decline stops.

Closing the Population

The inflow is a quantity, not a choice: U.S. immigration is rationed, so marginal changes in the value of living here do not move it. R and M are pinned at the baseline by two observables — the outside-origin share of young entrants (16.9%, ACS) and the population normalization — and held fixed in counterfactuals.

Population then responds to policy through births alone. Differentiating the balance condition:

$$\% \Delta S = \frac{1 - s^{out}}{s^{out}} \% \Delta B_0 \approx 4.9 \times \% \Delta B_0.$$

Each immigrant household founds a family line that fades out at the native replacement rate; a policy that raises births makes every line last longer.

Two qualifications. Housing damps the response: a larger population raises rents, which discourages births. And an earlier specification in which entrants weighed home values against an outside option is retired: its entry elasticity is not identified (appendix).

Policy Results

Funded Property-Tax and Grant Reforms

Raise the property tax from 1% to 2%; rebate lump-sum, or fund a purchase grant for family-sized homes. All changes vs. the funded 1% baseline, percent:

Arm	Policy	TFR	Births	Househ.	$-\Delta\text{Imm.}$	Price	Rent
Floor	2% tax	+0.38	+0.33	+1.70	1.74	-17.5	-6.1
Floor	tax + grant	+0.75	+0.67	+3.46	3.52	-17.3	-5.8
Tilt	2% tax	+0.20	+0.18	+0.90	0.92	-17.7	-6.3
Tilt	tax + grant	+0.49	+0.44	+2.23	2.26	-17.2	-5.7

In U.S. units, +0.33–0.67% of births is roughly 12–24 thousand additional babies per year. The tax lowers the asset price, so the down payment falls by 17.5%, while renter rents — which include the tax — fall by 6%.

Reading the Population Numbers

The same number can be read three ways:

- more households in the long run, at a fixed immigration inflow;
- fewer immigrants required to hold the population fixed;
- the share of the fertility gap closed by the policy.

Housing feedback is present but small: the demographic arithmetic overstates the response by at most a tenth of a percentage point.

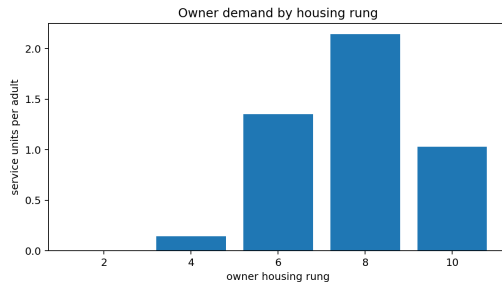
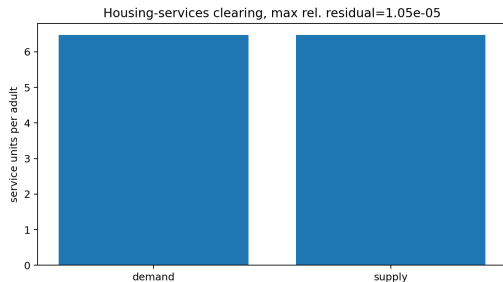
Timing: the birth and immigration flows move immediately, while the household count converges over generations — half of the adjustment takes roughly a century.

Open Decisions

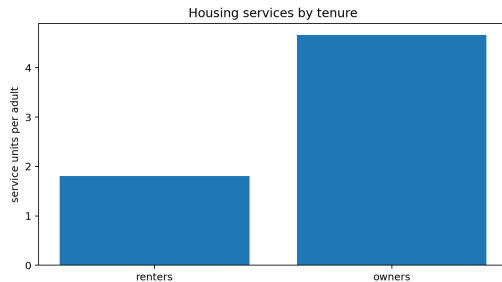
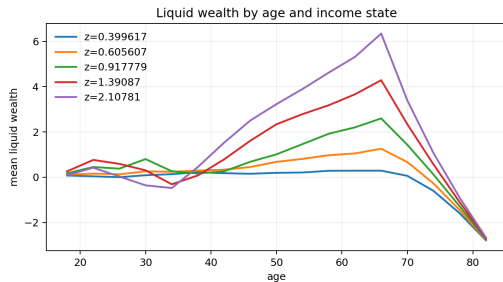
1. **Anchoring the inflow.** Among household heads aged 18–22, recent immigrants are only 3.7% — but immigrants mostly arrive older than the model's entry age. Either anchor the inflow to total net migration in entrant equivalents (roughly 17–22%, close to the 16.9% in use), or model entry at several ages
2. **The fertility block.** The closure requires natives to replace at least 83% of each cohort; the current fit sits close to that edge, and the childlessness and old-age wealth misses are limitations of the mechanism rather than of the search
3. **Welfare.** With different populations across counterfactuals, welfare is compared per household

Appendix

Appendix: Housing Market Diagnostics

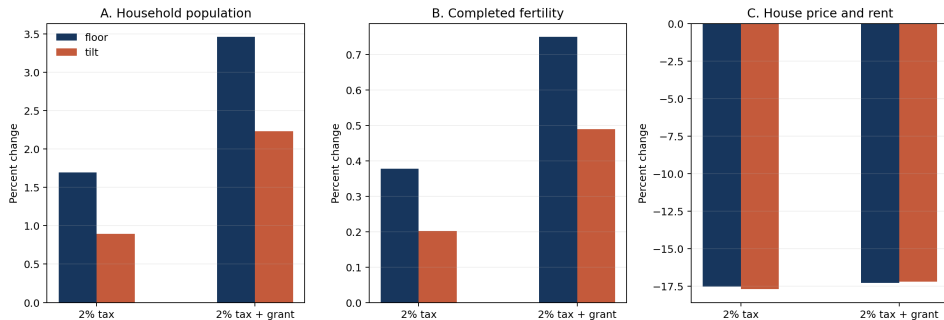


Appendix: Wealth and Tenure Diagnostics

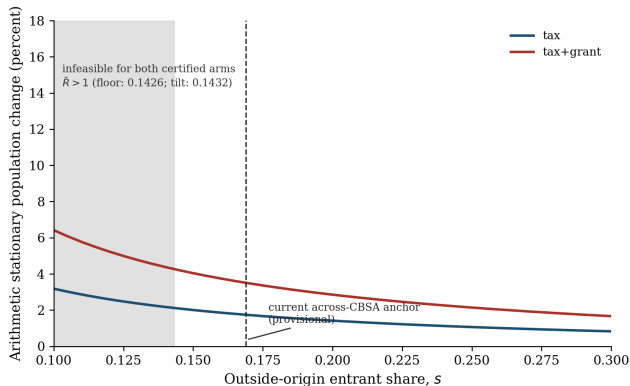


Appendix: Policy Comparison

Experimental repaired-E5 funded policies: quota population closure



Appendix: Anchor Sensitivity and Convergence Speed



Stationary population response as a function of the outside-origin share s^{out} , with the infeasible region ($s^{out} < 0.143$) shaded. Convergence: generational decay $\rho = \bar{R}B_0/E_0 = 0.831$ gives a half-life of 97–112 years ($g = 26$ –30); an age-structured variant with uniform birth timing gives ~ 230 years (upper bound).

Appendix: The Retired Logit as a Sensitivity Row

The old protocol's numbers survive as a labeled robustness row, not a model:

Arm	Policy	Quota population	Logit sensitivity ($\lambda = 2$, unidentified)
Floor	2% tax	+1.70	+9.98
Floor	tax + grant	+3.46	+10.10
Tilt	2% tax	+0.90	+8.99
Tilt	tax + grant	+2.23	+9.06

The gap between columns is what an unidentified taste parameter was contributing. The logit row bounds “what if entry responded to values” without pretending to identify it.

Appendix: National Entry-Anchor Measurement (ACS 2012–2023)

Measure (household heads unless noted)	Share	Multiplier $(1 - s)/s$
Ages 18–22, foreign-born, arrived $\leq 4y$ (headline)	3.71%	25.9
Ages 18–22, foreign-born, arrived $\leq 8y$	5.36%	17.7
Ages 18–25, foreign-born, arrived $\leq 4y$	3.47%	27.9
Ages 18–22, foreign-born stock (upper bound)	9.04%	10.1
Persons 18–22, foreign-born, arrived $\leq 4y$	2.89%	33.7
Quota feasibility floor	14.3%	—
Age-aggregated net-migration bracket (to verify)	17–22%	3.5–4.9

Every entry-age-window measure is infeasible; the age-aggregated concept is not. The two are different objects because immigrants mostly arrive above the stylized entry age — the anchor decision on the Open Decisions slide.

Appendix: Multi-Start Record (overnight)

16 fresh chains, unchanged 12-moment / 9-parameter contract, 225-minute budget each, strict-repeat certification:

	Best	Median	Worst
Fresh chains (seeds 09–24)	263.5	~284	370.5
Certified incumbent (Aug 5, chain 6)	249.186	— stands	

All 16 eligible with exact strict repeats on the run's best; market residuals $\leq 2.2 \times 10^{-5}$. With 24 chains total across the two arrays, the remaining misses are search-converged.